

AQA Level 3 Certificate — Mathematical Studies (1350)**End of Topic Test — Chapter 6: Confidence Intervals (Test 2)***Point estimates, sampling distributions and confidence intervals*

Name: Class: Date:

Time: 50–60 minutes

Total marks: 48

Instructions

- Answer all questions in the spaces provided.
 - Show all necessary working; otherwise marks for method may be lost.
 - Give clear interpretations in context where requested.
 - You may use a calculator.
-

1. Confidence interval for a mean

A sports scientist measures the reaction time (in milliseconds) of a random sample of $n = 20$ athletes. The sample mean is $\bar{x} = 215.6$ ms and the sample standard deviation is $s = 18.4$ ms.

- (a) Calculate a 95% confidence interval for the population mean reaction time. Use the t -distribution and give your answer to one decimal place. Show all steps, including the degrees of freedom and the critical t -value you use. [5]
- (b) Interpret the interval in context. State clearly what the 95% confidence level means. [3]
- (c) Explain briefly why the t -distribution is used here rather than the Normal z -distribution. [2]

2. Confidence interval for a proportion

A retailer surveys a random sample of $n = 500$ customers about whether they prefer a new packaging design. Of these, 310 customers say they prefer it.

- (a) Calculate a 95% confidence interval for the true proportion of customers who prefer the new design. Give your answer as percentages to one decimal place. [4]

- (b) The retailer's marketing team claims that "at least 55% of customers prefer the new design". Using your interval, assess whether this claim is supported by the sample data. Explain your reasoning. [3]

- (c) State one condition that should be checked before using the Normal approximation for a proportion confidence interval, and verify whether it holds here. [2]

3. Sample size for a desired margin of error

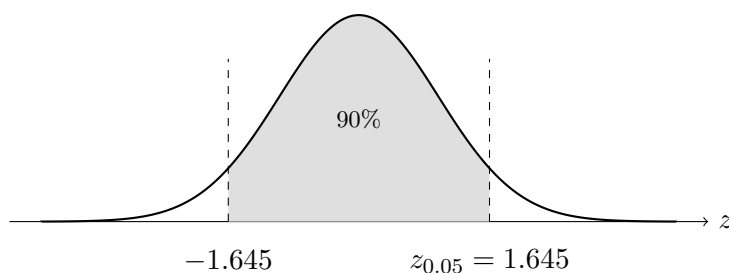
A delivery company wants a 95% confidence interval for the mean delivery time of parcels to have a margin of error no greater than 1.5 minutes. From a pilot study, the population standard deviation is estimated as $\sigma \approx 6.0$ minutes.

- (a) Calculate the minimum sample size required. Show your working and give the final n as a whole number. [4]

- (b) If the company instead wanted a 90% confidence level, explain qualitatively how the required sample size would change and why. [2]

4. Critical values and confidence levels

The diagram below shows a standard Normal curve with the central 90% region shaded.



- (a) Explain how the critical value $z_{0.05} = 1.645$ is used when constructing a 90% confidence interval for a mean with known σ . [3]
- (b) State the critical value you would use for a 99% confidence interval (Normal approximation) and explain briefly how the interval width would change compared with the 90% interval. [2]

5. Interpretation and misconceptions

Answer the following short questions.

- (a) A student calculates a 99% confidence interval for a mean as $(45.3, 52.7)$. Write a correct sentence interpreting this interval for a non-technical audience. [3]
- (b) Explain why it is incorrect to say: “There is a 99% probability that the population mean lies between 45.3 and 52.7.” [3]
- (c) Give one practical reason why a researcher might prefer a *narrower* confidence interval. [2]

6. Applying confidence intervals to decision making

A bakery adjusts its recipe to reduce the mean sugar content of its cereal bars. A random sample of $n = 40$ bars produced under the new recipe has mean sugar content $\bar{x} = 18.4$ g and sample standard deviation $s = 2.4$ g. The bakery's original target maximum mean sugar content is 20 g.

- (a) Construct a 95% confidence interval for the true mean sugar content under the new recipe. Give your answer to two decimal places. [4]
- (b) Based on your interval, discuss whether there is evidence that the new recipe achieves a mean sugar content below the target of 20 g. Give a clear conclusion in context. [3]
- (c) The quality manager states that the new recipe qualifies for a “reduced sugar” label provided the upper bound of the 95% confidence interval does not exceed 19.5 g. Using your interval, evaluate this criterion and state whether the recipe meets it. [3]

END OF TEST